STACKS, EXAMPLES OF DEFORMATIONS: LEMMA-JOUANOLOU-TYPE-THING

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 3150-3166 of sources/stacks-examples-defos/pinned-original-source.tex; proof span is lines 3168-3213, the complete author proof (44 body lines).

Editorial note. Navigation stubs: 3 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 9 of queue_examples-defos.json (pb=1896, stmt 3150-3166, proof 3168-3213).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 054J: sources/stacks-examples-defos/pinned-original-source.tex, source lines 3213–3213. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 07C4: sources/stacks-examples-defos/pinned-original-source.tex, source lines 3213–3213. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 005K: sources/stacks-examples-defos/pinned-original-source.tex, source lines 3213–3213. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let k be a field. Let $1 \leq c \leq n$ be integers. Let $f_1, \ldots, f_c \in k[x_1, \ldots x_n]$ be elements. Let a_{ij}, $0 \leq i \leq n$, $1 \leq j \leq c$ be variables. Consider*

$$g_j = f_j + a_{0j} + a_{1j}x_1 + \ldots + a_{nj}x_n \in k[a_{ij}][x_1, \ldots, x_n]$$

Denote $Y \subset \mathbf{A}_k^{n+c(n+1)}$ the closed subscheme cut out by $g_1, \ldots, g_c$. Denote $\pi : Y \to \mathbf{A}_k^{c(n+1)}$ the projection onto the affine space with variables a_{ij}. Then there is a nonempty Zariski open of $\mathbf{A}_k^{c(n+1)}$ over which π is smooth.

Date: stacks-project:examples-defos.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. Recall that the set of points where π is smooth is open. Thus the complement, i.e., the singular locus, is closed. By Chevalley's theorem (in the form of Morphisms, Lemma 054J) the image of the singular locus is constructible. Hence if the generic point of $\mathbf{A}_k^{c(n+1)}$ is not in the image of the singular locus, then the lemma follows (by Topology, Lemma 005K for example). Thus we have to show there is no point $y \in Y$ where π is not smooth mapping to the generic point of $\mathbf{A}_k^{c(n+1)}$. Consider the matrix of partial derivatives

$$(\frac{\partial g_j}{\partial x_i}) = (\frac{\partial f_j}{\partial x_i} + a_{ij})$$

The image of this matrix in $\kappa(y)$ must have rank $< c$ since otherwise π would be smooth at y, see discussion in Smoothing Ring Maps, Section 07C4. Thus we can find $\lambda_1, \ldots, \lambda_c \in \kappa(y)$ not all zero such that the vector $(\lambda_1, \ldots, \lambda_c)$ is in the kernel of this matrix. After renumbering we may assume $\lambda_1 \neq 0$. Dividing by λ_1 we may assume our vector has the form $(1, \lambda_2, \ldots, \lambda_c)$. Then we obtain

$$a_{i1} = -\frac{\partial f_j}{\partial x_1} - \sum\nolimits_{j=2,\ldots,c} \lambda_j (\frac{\partial f_j}{\partial x_i} + a_{ij})$$

in $\kappa(y)$ for $i = 1, \ldots, n$. Moreover, since $y \in Y$ we also have

$$a_{0j} = -f_j - a_{1j}x_1 - \ldots - a_{nj}x_n$$

in $\kappa(y)$. This means that the subfield of $\kappa(y)$ generated by a_{ij} is contained in the subfield of $\kappa(y)$ generated by the images of $x_1, \ldots, x_n, \lambda_2, \ldots, \lambda_c$, and a_{ij} except for a_{i1} and a_{0j}. We count and we see that the transcendence degree of this is at most $c(n+1) - 1$. Hence y cannot map to the generic point as desired. $\square$